

IoT based Smart Water Quality Monitoring System

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Abstract- The Internet of Things (IoT) is the network of physical devices, vehicles, home appliances and other items embedded with electronics, sensors, actuators, software and connectivity which enables these things to connect and exchange data. Water quality is major environmental problem in India. The largest source of water quality in India is intreated sewage. Other sources of pollution include agricultural runoff and unregulated small-scale industry that results in polluting, most of the rivers, lakes and surface water. In conventional systems, the monitoring process involves the manual collection of sample water from various regions, followed by testing and analysis. The sensors connected with Arduino-uno in discrete way to detect the water parameters and to display in a desktop application. The quality of water should be monitored continuously, to ensure the safe supply of water from any water bodies and water resources for drinkable and using it and that continuous measurement of water condition based on the parameters IoT helps it in improving the health and living standards of all living things.

Keywords: pH sensor, Temeperature Sensor, Soil Moisture Sensor, Wi-Fi, Internet of Things (IoT).

I. INTRODUCTION

In the 21st century, there were lots of inventions, but at the same time were pollutions, global warming and so on are being formed, because of this there is no safe drinking water for the world's pollution. Nowadays, water quality monitoring in real time faces challenges because of global warming limited water resources, growing population, etc. Hence there is need of developing better methodologies to monitor the water quality parameters in real time[1]. The water quality parameters pH measures the concentration of hydrogen ions. It shows the water is acidic or alkaline. Pure water has 7pH value, less than 7pH has acidic, more than 7pH has alkaline. The range of pH is 0-14 pH. For drinking purpose it should be 6.5-8.5pH.

The impact of water on any living beings is beyond description. With the rapid increase of world population, water management becomes an important issue specially in industrial, agricultural and other sectors. Most of the people around the world lack behind drinkable water. Every year many people are suffering from various fatal diseases caused by water pollution. Research has found that around 5 million death is caused only because of drinking unsafe water. Research by WHO (World Health Organization) shows that almost 1.4 million of child death can be prevented by providing drinkable water to them. The primary objective of this project is to introduce an intelligent water quality monitoring system in IoT (Internet of Things) platform which would help to monitoring different physical parameters of the drinkable water rather than relying on

manual process. Moreover, IoT is a system of alliance among various devices and the competence of deportation data over the system. In modern societies proper management of wastewater is a necessity, not an option. pH is an important limiting chemical factor for aquatic life. If the water in a stream is too acidic or basic, the H^+ or OH^- ion activity may disrupt aquatic organisms' biochemical reactions by either harming or killing the stream organisms.

Internet of Things (IoT) is the combination of a physical object, controller, sensor, actuators, and Internet. IoT plays vital role in Industry 4.0. Actuators and sensors are the devices that interact with real-world physical objects and collect data from the environment. Actuators and sensors, process the electrical input and transform into tangible action. IoT devices collect a huge amount of data, that are valuable and useful which should be stored, organized and processed.

Therefore a continuous series of anomalous measurements would indicate the potential introduction of a water pollutant and the user will be notified of this activity with the aid of IoT technology. The goal of this research is to develop a IoT based Smart Water Quality Monitoring System. physical parameters: temperature, pH, conductivity and temperature of different water samples are measured via these separate sensors equipped with Arduino Uno and NodeMCU.

A desktop application is developed in .NET platform to identify whether the tested water samples are safe or unsafe for human consumption.

II. LITERATURE REVIEW

water quality monitoring methods, sensors, embedded design, and information dissipation procedure, role of government, network operator and villagers in ensuring proper information dissipation While automatically improving the water quality is not feasible at this point, efficient use of technology and economic practices can help improve water quality and awareness among people. To ensure the safe supply of drinking water the quality should be monitored in real time for that purpose new approach IOT (Internet of Things) based water quality monitoring system has been proposed. In this paper, we present the design of IOT based water quality monitoring system that monitor the quality of water in real time. This system consists some sensors which measure the water quality parameter such as pH, temperature, soil moisture. The measured values from the sensors are processed by microcontroller and this processed values are transmitted remotely to the core controller that is NodMCU. Finally, sensors data can view on internet browser.

III. SYSTEM OVERVIEW

The proposed IoT based Smart Water Quality Monitoring System is able to read data from water samples by sensors through the microcontroller and NodeMCU these values will be displayed on the web browser. The proposed block diagram of based Smart Water Quality Monitoring System consists of different sensors connected with controller and NodeMCU to measure important physical parameters (pH, temperature, soil moisture) of water samples. The pH sensor is used to measure the presence of acidity or alkalinity of any solution in logarithmic scale. The temperature sensor provides accurate reading between -55 to 125°C . To measure the water is presented or not. The recommended detection range of this sensor is 1 to 15 ms/cm within a temperature between 0 - 40°C . The extracted data from these sensors are accessed by the controller arduino-uno and transfer to NodeMCU and then to the developed web browser. The test water sample is "Drinkable" or "Not Drinkable",

IV. PRINCIPLE

The main principle of this circuit is that the device placed at any particular place can measure the quality of water based on pH scale and the temperature of water and soil moisture. These three different sensors plays a major role pH scale to measure the in the scale of 0 - 14 .and soil moisture sensor is used to measure inform of percentage of 0 - 100 percentage to check whether the water is being sent to the fields, tanks etc., and the

temperature is used to measure the temperature of water to provide to the plants, and with in the limit of temperature to drink the water.

These values are displayed on the web browser with the assist IoT technology

V. WORKING

There are two parts of programming in this Smart Water Quality Monitoring System using IoT. In the first part, Arduino UNO is programmed and in the second part, NodeMCU will be programmed.

The proposed IoT based Smart Water Quality Monitoring System the data from the pH value detects the value, temperature in the form of centigrade and soil moisture sensor in the form of percentage, the data from the sensors will be displayed on the web page in the form of tabular form which Is shown in figure 5.1

Parameters	Value	Units
PH Value	3	N/A
Temperature	31	Centigrade
Moisture	1	%

Fig 5.1 Expected Output from Web Browser

VI. MEASUREMENT PARAMETERS OF IOT BASED SMART WATER QUALITY MONITORING SYSTEM

Basically, there are many parameters that are needed to be measured for IoT based water quality Monitoring system. However, the system proposed measures the key water parameters:

- Water's pH value.
- Temperature.
- soil moisture.

VII. BLOCK DIAGRAM

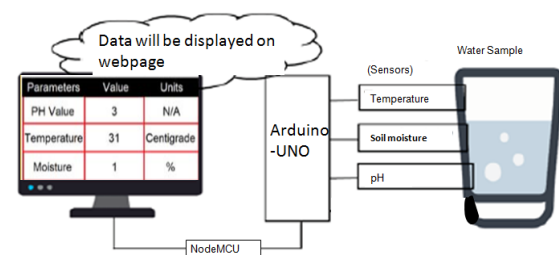


Fig 7.1 Block diagram of IoT based Smart Water Quality Monitoring System.

In this block diagram, there are three sensors are the major role with the help of IoT technology, it can be predicted it is drinkable water or not drinkable water. The sensors are temperature sensor, soil moisture sensor and pH sensor. These sensors play the vital role in this system.

1. Test for Pure Water

First step collect the water sample from the tank which is sensor and dipped in to it, the value will be displayed on to the web browser, so that the water sample taken by tank and tested, so the ph value with in the limit 7, so it can be drinkable, usable water to wash the eatable items.

2. Test for Salt water

Collecting the water sample from river, which is collected and the ph sensor dipped in to it, the data will be converted in to ph scale from the ph conversion board the data will be flowed in to through arduino uno and nodemcu and the ph value will be displayed and we can check the temperature, so that the temperature says how much centigrade its having the number.

3. Test for Plant Pot Water

Collecting the water from the plant pots and checking the sample with the help of the different sensors are soil moisture, pH and temperature and the water was tested and displayed it in web browser. soil moisture says the water is in terms of percentage presented and the temperature also checked to know the nutrients do not damaged in the plants.



Fig 7.2 Monitoring System of Water

VIII. TECHNOLOGY KEY TERMS

Arduino UNO: Arduino is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller. Arduino Software (IDE) were the reference versions of Arduino, now evolved to newer releases. The Uno board is the first in a series of USB Arduino boards, and the

reference model for the Arduino platform; for an extensive list of current, past or outdated boards see the Arduino index of boards.

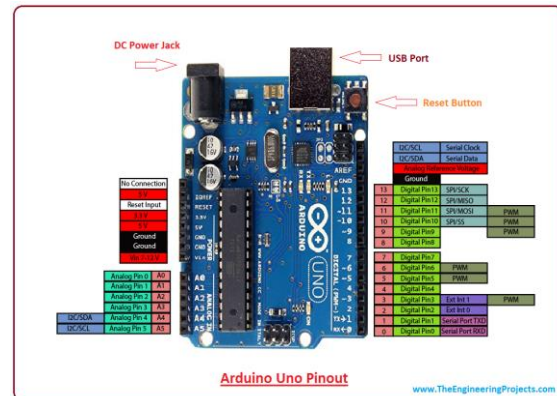


Fig 8.1 Pin Diagram of ArduinoUNO

NodeMCU Wifi module: The ESP8266 WiFi Module is a self contained SOC with integrated TCP/IP protocol stack that can give any microcontroller access to your WiFi network. The ESP8266 is capable of either hosting an application or offloading all Wi-Fi networking functions from another application processor. Each ESP8266 module comes pre-programmed with an AT command set firmware. The ESP8266 module is an extremely cost effective board with a huge, and ever growing, community

The ESP 8266 Wi-Fi module is a low cost component with which manufacturers are making wirelessly networkable microcontroller module. ESP 8266 WiFi module is a system-on-a-chip with capabilities for 2.4GHzrange. It employs a 32 bit RISC CPU running at 80 MHz. It is based on the TCP/IP (Transfer control protocol). It is the most important component in the system as it performs the IOT operation. It has 64 kb boot ROM, 64 kb instruction RAM, 96 kb data RAM. Wi-Fi unit performs IOT operation by sending energy meter data to webpage which can be accessed through IP address. The TX, RX pins are connected to the 7 and 8 pins of the Arduino microcontroller.

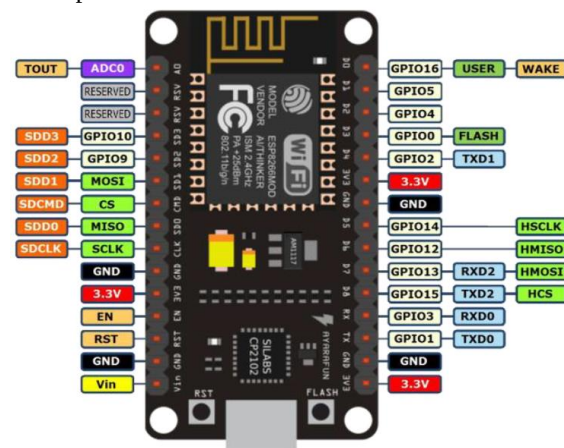


Fig 8.2 Pin diagram of NodeMCU ESP8266

pH sensor: The pH of a solution is the measure of the acidity or alkalinity of that solution. The pH scale is a logarithmic scale whose range is from 0-14 with a neutral point being 7. Values above 7 indicate a basic or alkaline solution and values below 7 would indicate an acidic solution. It operates on 5V power supply and it is easy to interface with arduino. The normal range of pH is 6 to 8.5.



Fig 8.3 pH Sensor

Temperature sensor: Water Temperature indicates how water is hot or cold. The range of DS18B20 temperature sensor is -55 to +125 °C. This temperature sensor is digital type which gives accurate reading.



Fig 8.4 Temperature Sensor

Soil moisture sensor: soil Moisture Sensor uses capacitance to measure dielectric permittivity of the surrounding medium. And it can act like a switch to this device

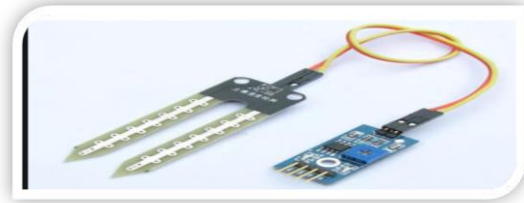


Fig 8.5 Soil Moisture Sensor

IX. CIRCUIT DIAGRAM AND SYSTEM DESIGN OF MODEL

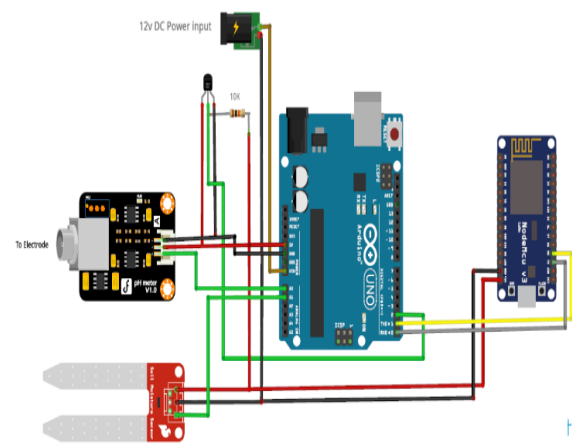


Fig 9.1 Circuit Diagram of IoT based Smart Water Quality Monitoring System

In this circuit diagram Arduino UNO is connected to transmitter and receiver in connected to NodeMCU, then the data of the sensors which are collected from the sensors are transmitted from the pH sensor, soil moisture sensor and Temperature sensor to connected NodeMCU.



Fig 9.2 Design Diagram of IoT based Smart Water Quality Monitoring System

The output pin of positive and negative, positive is connected to A0 in Arduino UNO and negative to the Ground to Ground in Arduino UNO. The input for the pH conversion board is 9v dc. And the pH sensor probe is connected to the pH conversion board. And the soil moisture sensor pins connected to chip of moisture positive to positive chip, and negative to negative chip, from the chip vin,grd,D0 is connected to Arduino UNO , vin to vcc, grd to grd, D0 to A1. The temperature sensor is positive is connected to A2 of Arduino UNO and negative is connected to ground of NodeMCU.

The output of these sensors sent to NodeMCU and then its connected to wifi server with the NodeMCU programming the html page is created, the data will be sent and displayed and at every 5 Seconds the web browser page will be refresh so the new data will displayed on screen.

XI. APPLICATIONS

- Agriculture fields
- Water purifiers at low cost
- Industries
- Home
- Hotels
- Airports
- Public Transportations stations etc.,

XII. ADVANTAGES

- Mostly used soil moisture sensor in agriculture fields so nutrients will be safe.
- By the pH value, it can able to know drinkable, usable, etc.,
- In industries mostly used temperature and pH sensors whether it is under temperature and quality water used in instant products
- More sensors and personal applications we can emerge....

XIII. DISADVANTAGES

- Day by day the device using the pH electrode may get damaged and it may not worked further....
- More programming should be done, it should be connected to internet daily...
- If it is placed in agriculture fields openly the device get damaged by weather, disturbing animals etc.,
- If we want to keep more sensors in that device at required place , the place will not be in that required device if it is short permanent circuit board.

XIV. CONCLUSION

Monitoring of soil moisture, PH & Temperature sensors of Water makes use of water detection sensor with unique advantage and existing network. The system can monitor water quality automatically, and it is low in cost and does not require people on duty. So the water quality testing is likely to be more economical, convenient and fast and displayed on screen. The system has good flexibility. Only by replacing the corresponding sensors and changing the relevant software programs, this system can be used to monitor other water quality parameters. The operation is simple. The system can be expanded to monitor industrial and agricultural production and so on. It has widespread application and extension value. By keeping the embedded devices in the environment for monitoring enables self protection (i.e., smart environment) to the environment. To implement this need to deploy the sensor devices in the environment for collecting the data and analysis. By deploying sensor devices in the environment, we can bring the environment into real life i.e. it can interact with other objects through the network. Then the collected data and results will be available to the end user through the Wi-Fi and the values displayed on screen.



Fig 14.1 IoT Technology using in Water Monitoring System

XV. FUTURE SCOPE

Detecting the more parameters for most secure purpose, Increase the parameters by addition of multiple sensors. Its major limitation is the accuracy. It can be improved by taking a large number of readings and displaying them. The ultimate goal of this work is to observe the quality of water samples by designing a IoT based smart water quality monitoring system implemented in that can detect specific physical parameters: Temperature, pH, Soil Moisture sensors. The future scope of this project is monitoring environmental conditions, drinking water quality, treatment and disinfection of waste water etc. This system could also be implemented in various industrial processes. The system can be modified according to the needs of the user and can be implemented along with lab view to monitor data on computers.



Fig 15.1 Water Monitoring System using IoT

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